

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf__

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _S0/0/0_

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _129_

Does R2 show up as an OSPF neighbor on R1? _Yes_

Does R2 show up as an OSPF neighbor on R3? _No_

What does this information tell you?

All traffic from R3 to the 192.168.2.0/24 network is being routed through R1. This is because interface S0/0/1 on R2 is still configured as a passive interface, so OSPF routing information is not advertised between R2 and R3 on that link. The accumulated cost of 129 comes from two serial links with cost 64 each plus the G0/0 interface on R2 with cost 1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_R2(config)# router ospf 1
R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _S0/0/1_

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65: Cost = 64 (Serial link) + 1 (Gigabit Ethernet)

Is R2 listed as an OSPF neighbor to R3? _Yes_

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Modern networks support link speeds higher than 100 Mb/s. The default OSPF reference bandwidth is 100 Mb/s, which may cause many high-speed interfaces to have the same cost. Increasing the reference bandwidth allows OSPF to calculate more accurate routing costs for faster links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24 is calculated from the sum of the outgoing interface costs:

R1 S0/0/1 (781) + R3 G0/0 (1) = 782.

The cost to 192.168.23.0/30 is calculated from R1 S0/0/1 (781) + R3 S0/0/1 (64) = 845.

OSPF calculates the total route cost by adding the costs of all outgoing interfaces along the path from the source router to the destination network.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new accumulated cost to the 192.168.23.0/24 network on R1 is 1562.

This is because the bandwidth of the serial interfaces is changed to 128 kb/s, which increases the OSPF cost of each serial link. The total route cost is the sum of the costs of the interfaces along the path.

Step 3: Change the route cost. c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the path with the lowest accumulated cost.

After the bandwidth changes, the cost of the path through R2 becomes lower than the direct path.

The cost through R2 is: R1 S0/0/0 + R2 S0/0/1 + R3 G0/0 = 781 + 781 + 1 = 1563

The direct path cost is: R1 S0/0/1 + R3 G0/0 = 1565 + 1 = 1566

Since 1563 < 1566, OSPF chooses the route through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?
Router ID uniquely identifies each router in the OSPF network and is used in DR/BDR elections. Setting it manually ensures stability and prevents it from changing if an interface goes down.
2. Why is the DR/BDR election process not a concern in this lab?
Because the network uses point-to-point serial links. DR/BDR elections only occur on multi-access networks such as Ethernet.
3. Why would you want to set an OSPF interface to passive?
A passive interface prevents OSPF neighbor formation and unnecessary routing updates on that interface while still advertising the network to other routers.
4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?
OSPF uses a hierarchical design with multiple areas connected through the backbone Area 0. This area structure reduces routing table size and limits routing updates, allowing the network to scale efficiently.
5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?
network 10.0.0.0 0.255.255.255 area 0
6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?
The OSPF route will be used. This is because OSPF has the lowest administrative distance (110) compared with RIP (120) and EIGRP external (170). The route with the lowest administrative distance is selected for routing.